Database relationships — how the rows actually connect

Full ER sketch: PROJECT_KNOWLEDGE_MAP §6. Yahan the three spines, row-level:

Spine 1 — material

ClothRoll.adda (FK, PROTECT-ish semantics via guards): ek roll EK hi Adda ka. Leftover row = (roll, source_adda, weight) — consumption ka exact fact. Example: `CR-000142 → 3-PATTI-001, 25.40kg attach, leftover 3.10kg` ⇒ consumed = 22.30kg. G1 isi se cloth cost nikalega — retroactively.

Spine 2 — work (production truth)

Adda → AddaStageRecord (one per WorkflowStage) → WorkerStageTask (per worker) → WorkerStageContribution (per color/size line). Row: `WSC(task=utest@cutting, color=Red, size=S1, reported=60, verified=NULL, expected_rate=3.00 frozen, settlement_line=SWA#42)`.

Spine 3 — money (financial truth)

AddaSettlement → SWA earning lines → WorkerLedgerEntry credits; PSI recovery rows → ledger debits; frozen AddaSettlementItem per worker. The provenance loop: WSC.settlement_line → SWA.adda_settlement → ADST; ledger.assignment → SWA. Har paise ka raasta dono taraf walk hota hai.

Why JOIN-walks stay cheap

select_related (FK joins) + prefetch_related (reverse sets) at the read layer (payroll_service); indexes on the hot pairs (worker+status, adda+settled_at, task). Perf baselines pin query counts in tests — N+1 regression = failing test, silent nahi.